Digital-Twin-Oriented Pedagogical Integration of BatchSep for Dynamic Modeling and Validation of MeOH/H_2O Batch Distillation

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ABSTRACT

Despite growing interest in digital twin applications in engineering education, a critical gap remains in process system engineering (PSE) pedagogy. Existing approaches either use digital twins as passive visualization tools disconnected from the dynamic behavior of physical systems, implementations rely solely on high-fidelity simulators without real-time cyber-physical integration, or proposed frameworks remain conceptual without pilot-scale validation. Consequently, the full pedagogical potential of digital twin technology in chemical engineering education has yet to be realized. To date, no previous study has described a fully operational, closed-loop digital twin architecture specifically designed for batch distillation (BD) education that integrates a physical unit with dynamic simulation, advanced control, and real-time data exchange within unified platform.

To overcome this limitation, a four-layer cyber-physical framework was established and deployed. The system combines a pilot-scale with Aspen BatchSep dynamic modeling, MATLAB/Simulink-based control and state estimation, and a LabVIEW human-machine interface (HMI) / supervisory control and data acquisition (SCADA) interface interconnected via open platform communications unified architecture (OPC-UA) and message queuing telemetry transport (MQTT) protocols for real-time communication. Experimental validation was performed by comparing measured process data with simulation predictions and tuning control strategies using actual batch operation data. The resulting platform provides a reproducible and experimentally validated educational framework that supports the progressive development of competencies in instrumentation, model validation, and closed-loop digital twin deployment aligned with Industry 4.0 requirements for modern PSE graduates.

Keywords: Chemical engineering education; Batch distillation; digital twin education; Aspen BatchSep pedagogy

1. INTRODUCTION

Laboratory instruction in chemical unit operations has long followed a teacher-directed model in which theoretical content is conveyed through formal lectures, collectively structured exercises, and written examinations drawn predominantly from established textbook material [1]. However, despite its widespread use, this instructional paradigm presents several well-documented shortcomings. Laboratory sessions remain highly structured, affording minimal opportunity for independent judgment or genuine analytical reasoning [2]; and graduates enter professional practice with limited exposure to the digital technologies that now underpin industrial chemical engineering [3]. These deficiencies point toward the necessity of adopting learner-centered pedagogical strategies that bridge computational modelling, hands-on experimentation, and real-time industrial data environments [4]. The deliberate incorporation of advanced process simulation platforms-notably Aspen PLUS and related equation-oriented environments-has been demonstrated to yield measurable gains in student learning outcomes

by enabling parametric investigation, inquiry-based learning, and design-of-experiment (DoE) reasoning prior to physical laboratory implementation [5]. Such approaches cultivate deeper conceptual understanding and actively promote critical analytical reasoning [2].

Extending these benefits, the present study embeds rigorous process simulation within a digital twin-oriented educational framework that couples dynamic modelling tools, specifically Aspen BatchSep and MATLAB/Simulink, with live experimental validation and industrial cyber-physical infrastructure within a unified cohesive laboratory ecosystem [6].
Within this integrated environment, the laboratory is reoriented from static, procedure-driven exercise into a dynamic, data-informed activity that equips graduates with Industry 4.0 PSE competencies [7].

2. METHODOLOGY FOR THE PEDAGOGICAL FRAMEWORK

As presented in Table 1, the digital twin architecture comprises four distinct functional layers governing bidirectional data flow between the physical batch distillation column and its computational twin, integrating theoretical understanding, rigorous process modelling, experimental validation, and industrial digital technologies within a unified learning ecosystem aligned with ABET accreditation standards. The three-pillar pedagogical structure deliberately transforms a passive boil-and-collect laboratory exercise into a comprehensive, inquiry-driven engineering task.

Pillar 1 establishes the thermodynamic and safety foundation prerequisite to physical plant operation. Students construct vapor-liquid equilibrium (VLE) models for binary systems, including methanol/ethanol-water mixtures, consider the viability of relative volatility (α), generating T_{xy} and xy phase diagrams, applying the Rayleigh equation and McCabe-Thiele graphical method to predict batch distillation dynamics, and distinguishing azeotropic behavior across multicomponent systems [5]. Industrial safety culture is embedded through structured HAZOP analysis and PPE protocol implementation, complemented by a hands-on technical audit of the pilot plant electrical infrastructure encompassing single-line diagram interpretation, input/output signal architecture, and feedback control loop examination on Piping and Instrument Diagram (P&ID). Systematic evaluation of constant and variable reflex control strategies [8] progressively transitions students from passive observers to active decision-makers, grounded in data-driven reasoning and operationally justified through a rigorous license-to-operate framework aligned with contemporary industrial practice.

Pillar 2 transitions the laboratory from theoretical understanding to physical pilot-scale operation, wherein students evolve from passive learners into plant operators managing near-industrial dual distillation columns under real-time process monitoring. Troubleshooting of non-ideal column behaviors, including flooding, weeping, entrainment, and pressure drop variations [9], students develop the analytical rigour and practical decision-making capability required to manage complex, uncertain industrial processes.

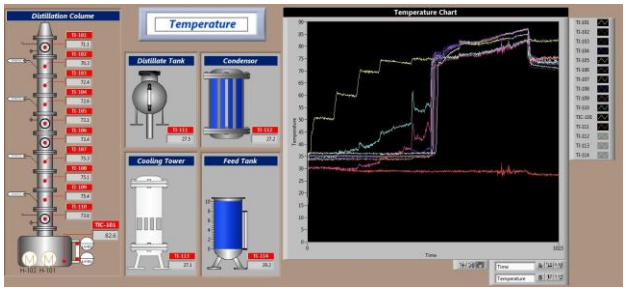
Pillar 3 wherein students transition from plant operators to proactive process engineers tasked with constructing and validating a digital twin of the BD column within Aspen Batch Modeler. Students rigorously characterize the as-built column by incorporating holdup, tray and packing details, and height equivalent to a theoretical plate (HETP), while thermodynamic model selection between NRTL and UNIQUAC is justified on the basis of mixture non-ideality [10]. Model fidelity is established by reconciling experimental and simulated profiles through equilibrium and rate-based approaches, performing sensitivity analysis on key separation parameters, and achieving a validated base case within an Average Absolute Deviation (AAD%) criterion of ±5%. Dynamic visualization of shifting McCabe-Thiele operating lines as the still pot depletes provides an intuitive, real-time reinforcement of equilibrium theory inaccessible through conventional static instruction.

3. PILOT PLANT APPARATUS AND SIMULATION FRAMEWORK

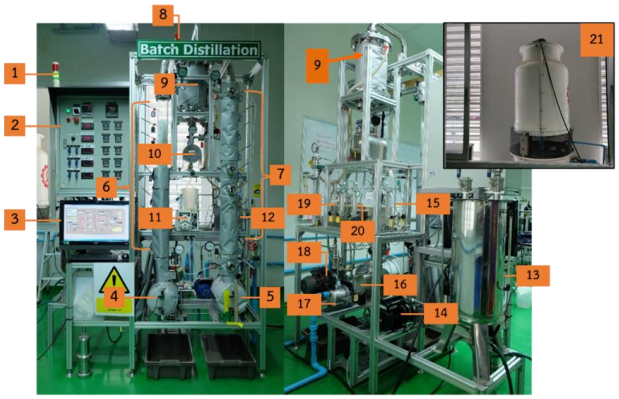
As displayed in Figure 1b, a self-swap tray-packed distillation column, students can be switching a column required with dual functionality, allowing it to switch between two distinct internal configurations as sieve-tray operation and packed-column operation. The system features an SS304L stainless-steel column with a 10 cm inner diameter and 182 cm total height, divided into 11 modular sections arranged bottom-up, with stage 1 housing the reboiler. Three sections incorporate side glass ports with five viewing windows each for real-time visual monitoring. The internal column operates under atmospheric or vacuum conditions, with ceramic wool insulation and silica-fabric cladding minimizing heat loss throughout operation. Students can select various feed positions at three locations: mid-column (stage 6), near the bottom (stage 2), or at the top (stage 11) as an external reflux stream. For the feed-transfer section, the storage tank (13) supplies the fresh mixture through the feed pump (14), while the feed rotameter (15) provides a visual indication during charging. For safe automatic operation, the feed pump is interlocked to stop when either the reboiler high-level alarm (LAH) or limit switch is activated, indicating that the reboiler has reached its allowable filling level, or when the storage-tank low-level alarm (LAL) is detected. This control logic prevents reboiler overfilling and protects the feed pump from dry running.

Table 1. Pedagogical evidence mapping of the four-layer bidirectional synchronization loop-modern engineering tool usage (e.g., Aspen BatchSep, MATLAB/Simulink, OPC-UA integration, and real-time data analytics platforms) is primarily aligned with ABET student outcome (SO) 1, while also supporting outcomes (SO) 2 and 6 through design-oriented simulation and data-driven experimentation within the digital twin framework.

Layer	Component	Function	Protocol / Tool	Student learning activity
Layer 4	LabVIEW HMI	SCADA visualization, Virtual Instrument (VI) panels, real-time waveform charts	OPC-UA client	- reads and interprets live waveform charts, tray temperature trends, and composition profiles on LabVIEW VI - prepares written commentary on batch distillation progress for lab report
Layer 3a	MATLAB / Simulink	Kalman state estimator, PID/MPC control synthesis	MQTT / TCP socket	- formulates PID control strategy in MATLAB/Simulink, tuning K_p, T_i, and T_d before deploying heater duty (kW) and reflux valve setpoints (mL/min) via Node-RED. - validated against the Aspen BatchSep digital twin prediction to fulfill ABET SO5 and SO6. - compares Simulink-estimated tray compositions with Aspen BatchSep VLE profiles and adjusts thermodynamic package if AAD% > ±5% - calculates AAD% between model and simulation
Layer 3b	Aspen BatchSep	Rigorous VLE simulation, batch schedule optimization	REST API / CSV exchange	
Layer 2a	Raspberry Pi 5 + Node-RED	Edge gateway, Modbus master, MQTT publisher	USB-RS485 port, Modbus RTU and MQTT	- implements and validates the engineering unit scaling function within a Node-RED Function node - interrogates the live MQTT data stream using MQTT Explorer verifying the structured JSON payload structure and confirming that all nine process variables as (i) tray temperatures, (ii) column pressure, (iii) reflex and (iv) distillate flow rates that are correctly published to the designated topic at 100 ms - configures AI Utilities register map - sets engineering unit scaling for K-type thermocouple and 4-20 mA channels and validates calibration against known reference temp.
Layer 2b	RS485 serial Bus	9600 baud rate, Even parity, Data Bits: 8 and Stop Bit: 1	Modbus RTU	
Layer 2c	WISCO I/O + AI Utilities	Analog input (AI) conditioning, register mapping, Modbus slave	AI Utilities software configuration	
Layer 1	Batch distillation field Instruments	Thermocouples (TC), pressure transducers, flow meters, actuators (pump and heater) and solenoid valves	K-type TC, 4-20 mA, and digital I/O signals	- interpret batch distillation process and instrumentation diagrams using ISA 5.1 (P&ID) symbology - correctly identify instrument tag codes - trace each physical signal from field instruments through the digital I/O signals map to the Node-RED data acquisition pipeline



(a)



(b)

Figure 1 Pilot-scale batch distillation column utilized in the project-based unit operations laboratory. (a) LabVIEW HMI linked as SCADA visualization layer (b) the system integrates both packed and tray column configurations with supporting instrumentation and auxiliary equipment: (1) stack light (status indicator), (2) control panel, (3) all-in-one personal computer, (4) reboiler for packed column, (5) reboiler for tray column, (6) temperature indicator for packed column, (7) temperature indicator for tray column, (8) manual butterfly valve, (9) condenser, (10) reflux distillate tank, (11) peristaltic pump, (12) distillate sampling tank, (13) storage tank, (14) feed pump, (15) feed rotameter, (16) bottom tank, (17) cooling pump, (18) vacuum pump, (19) distillate rotameter, (20) reflux rotameter, and (21) cooling tower, respectively.

This reflux represents the lighter phase recovered from the decanter tank and recirculated to the column via a peristaltic pump (No. 11). A plate-and-shell heat exchanger (No. 9), mounted above the column, fully condenses the overhead vapor before cooling it to near-ambient temperature using 25 °C circulating cooling water, thereby minimizing MeOH evaporation during sample collection. Condensate subsequently flows to a knockout drum (No. 10), where the stream is divided according to the selected reflux or distillate flow rate. The liquid level in a receiver (No. 10) can be either automatically or manually regulated given its relatively aqueous-phase holdup capacity of approximately 5 L.

During heat-up ramping of the reboiler, operation is controlled by an automatic level-protection interlock employing an electromagnetic pulse-sensing probe with a coaxial cable, When the liquid level drops below the preset setpoint, the probe sends a signal to shut off the electric heaters (No. 4 and 5), thereby preventing overheating and helping maintain a stable vaporization rate. Two control strategies govern overall column operation. A PID controller regulates the temperature at stage 11 by adjusting reboiler duty as the manipulated variable, while an ON/OFF solenoid valve, driven by a 15-second reflux-ratio timer, manages reflux splitting. Under this timed sequence, condensate alternates between the distillate receiver and the column return line at equal intervals. When the timer is disabled, all condensate is returned to the column, representing total reflux conditions and ensuring consistent distillate production throughout each batch run.

To complement experimental observations and validate results under fewer simplifying assumptions, rigorous simulations are performed using the batch module in commercial Aspen PLUS Batch Modeler™ V14. The model is developed under standard assumptions, including negligible vapor holdup, adiabatic operation, perfect liquid-phase mixing, constant tray efficiency, atmospheric pressure, total condensation without subcooling, and phase equilibrium throughout. Non-ideal liquid-phase behavior is captured using the NRTL/UNIQUAC activity coefficient model, with binary interaction parameters obtained by regressed of VLE data for the MeOH/H_2O system.

The batch pot is initially charged with 8 kg of a binary mixture containing 50 wt.% MeOH in H_2O, with the column configured for 10 theoretical stages and a reboiler duty of 3 kW, corresponding to an approximate boilup rate of 0.25 kg/min. The column is first operated under total reflux for 20 minutes, after which a proportional reflux policy is introduced. The design specifications are summarized in Table 2 and Figure 2 illustrate a dynamic simulation of batch distillation column configured in Aspen BatchSep.

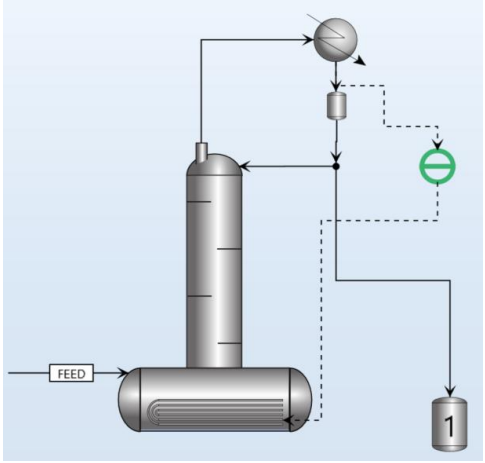


Figure 2 Dynamic batch distillation flowsheet modeled in Aspen BatchSep.

Table 2 Design specifications for Aspen BatchSep dynamic predictions.

Configuration	
Number of stages	12
Condenser	Total condenser
Initial condition	Total reflux
Pressure & Holdups	
Condenser pressure	1 atm
Reflux drum liquid holdup	1 kg
Pot (Reboiler)	
Volume	10 L
Diameter	20 cm
Reflux	
Reflux ratio	1

4. TEACHING AND LEARNING OUTCOMES AND CONCLUSIONS

From a pedagogical standpoint, the reboiler is first charged with an aqueous solution containing 50 wt. % MeOH, providing students with a well-defined binary feed composition as a starting bubble point on the T_{xy} diagram. Once the fresh feed is loaded, the condenser cooling system is activated before heating begins. An important procedural sequence that students must recognize is that initiating heating prior to activating the condenser can lead to uncontrolled vapor losses from the overhead system. The reboiler heating system is then switched on and the data acquisition system is initiated simultaneously. On the SCADA interface, students observe how the PID controller continuously adjusts the heater duty (kW) through incremental step changes (e.g., ramping up by +5 °C setpoint shifts) to regulate the mixture temperature. This dynamic response clearly illustrates the practical role of feedback control in maintaining stable and controlled column during transient startup. As the column progresses from nearing towards steady state, students are encouraged to track the temperature profiles across all stages. As shown in Figure 1a, students actively engage with the operational sequence and its corresponding process responses to develop a deeper understanding of BD dynamics the experiment begins with heating the reboiler under total reflux conditions, allowing the column to establish steady-state equilibrium. During this phase, no distillate being withdrawn. Until the reflux ratio (R = L/D) approaches infinity, the reboiler temperature is stabilized at approximately 95 °C, students can observe the stabilization of temperature gradients and internal vapor-liquid interactions, reinforcing theoretical concepts of phase equilibrium and column initialization.

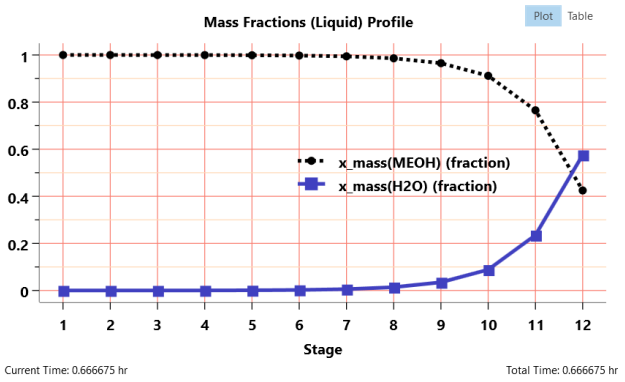
Students experimentally verified that under total reflux operation, the temperature across all trays in the column remains essentially constant, indicating the establishment of vapor-liquid equilibrium throughout the system. students systematically monitor and interpret key process variables, including (i) distillate flowrate profiles to evaluate production rates and transient behavior (ii) temperature profiles along the column to understand thermal gradients and stage-wise separation and (iii) liquid-phase composition profiles to assess both separation efficiency and main component distribution. To quantify composition, students collect liquid samples at designated points and analyze them using a Densitometer model D4. This hands-on analytical step strengthens the connection between measured physical properties and underlying thermodynamic principles.

At tray 1 (Top of the column), the distillate temperature remained approximately 65 °C for the first 25 minutes, corresponding to the boiling point of MeOH and indicating the collection of nearly pure light key component. Beyond this period, the temperature gradually increased toward 102 °C, signaling the onset of water-rich mixtures in the distillate. This transition provided a clear, real-time demonstration of composition drift in BD, allowing students to connect T_{xy} on Aspen PLUS properties with observable process behavior.

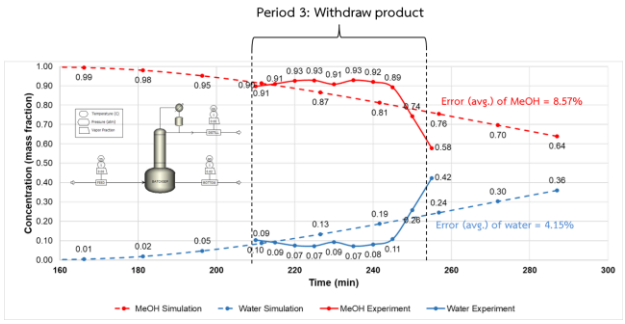
Figure 3a shows the distribution of liquid mass fractions of methanol and water across the column stages at total operation time of 40 min. Methanol and water system have high relative volatility ($\alpha > 1$), resulting effective separation in distillation. This profile represents a baseline operating condition provided to students for analysis and exploration, serving as a reference case to encourage critical thinking on process performance under modified conditions, for example reduced operation time or change the number of column stages, and their corresponding impact on separation efficiency.

As illustrated in Figure 3b, students compared the dynamic composition profiles of MeOH/H_2O obtained from Aspen BatchSep simulation and experimental measurements. The results demonstrate a strong qualitative agreement between the mathematical model and the physical system, particularly in capturing the key transition from MeOH-rich to H_2O-rich distillate over time. In the early stage of operation, students observed that the distillate is dominated by MeOH, with mole fractions approaching unity, consistent with its lower boiling point. As the batch progresses, the MeOH composition gradually decreases while the H_2O composition correspondingly increases, reflecting the depletion of the more volatile component in the reboiler. This transition is clearly captured by both simulation and experiment, enabling students to visualize the time-dependent nature of separation in batch distillation. Quantitatively, small deviations between simulation and experiment were observed, with an average error of approximately 8.57% for MeOH and 4.15% for H_2O. These discrepancies

provided valuable learning opportunities, prompting students to critically evaluate potential sources of error, including assumptions in thermodynamic models (e.g., VLE), non-ideal tray efficiency, heat losses, and measurement uncertainty.



(a)



(b)

Figure 3 (a) liquid-phase composition profile of methanol and water along the batch distillation column from Aspen BatchSep simulation (b) comparison of simulated and experimental distillate composition profiles for methanol and water during batch distillation, illustrating the agreement between Aspen BatchSep predictions and laboratory data, along with associated model deviations. (Reflux ratio = 1, total reflux time = 75 min, experimental heat duty (Q) = 466.7 W).

As shown in Figure 3b, this comparison plays a crucial role in bridging theory and practice. Students move beyond passive use of simulation tools to actively validate models against real experimental data, fostering skills in model verification, data interpretation, and critical thinking. Moreover, the ability of Aspen BatchSep to reproduce the overall trends reinforces its effectiveness as a digital twin tool for predicting batch distillation behavior. Overall, the exercise enables students to understand not only how composition evolves over time, but also how simulation and experiment can be integrated to support evidence-based decision-making in process design and operation.

The hands-on project tasks students with designing a batch distillation (BD) column for separating a MeOH/H_2O system and to determine an optimal reflux ratio for operation. In accordance with the assessment framework outlined in Table 3, the initial stage emphasizes conceptual design and process optimization, aligned with Kolb's experiential learning cycle [11] . During this phase, students draw upon prior knowledge while engaging in structured pre- to post-experimental tasks supported by simulation tools such as Aspen BatchSep and MATLAB/Simulink. This integrated approach enables the development of a theoretically grounded design strategy, including the selection of operating conditions, formulation of mass and energy balances, and prediction of system behavior under varying reflux ratios. From an ABET perspective, this stage supports SO6 through the effective use of modern engineering tools, while also reinforcing hypothesis-driven problem solving within a DOE framework, thereby preparing students for subsequent experimental validation and iterative refinement of their designs.

Survey data on the digital twin-oriented laboratory were collected over three successive academic years (2020-2021, 2021-2022, and 2022-2023). At the conclusion of each cycle, students were invited to complete an anonymous questionnaire consisting of the items listed in Table 4. The instrument was designed to evaluate the extent to which computer-based laboratory sessions and associated tutorials enhanced students' understanding of batch distillation processes and their proficiency with Aspen BatchSep and MATLAB/Simulink. Students also assessed the appropriateness of the weighting assigned to these practical components within the overall course grade, as well as their overall satisfaction with the instructional approach.

Responses were measured using a five-point Likert scale (1 = strongly disagree; 5 = strongly agree). Overall, the findings indicate consistently positive outcomes across all cohorts. Students widely agreed that the integration of Aspen BatchSep and MATLAB/Simulink significantly improved their conceptual understanding and technical skills. The scheduling and structure of the digital laboratory sessions were also regarded as appropriate, and the tutorials were identified as valuable in reinforcing software competency. Moreover, students expressed strong support for the continuation and expansion of such sessions to other advanced separation topics. Notably, while overall satisfaction remained high, some students indicated that the workload associated with post-laboratory assignments was substantial and suggested that greater weighting within the course assessment would more accurately reflect the effort required. in response, a new module on supercritical fractionation using Aspen BatchSep and MATLAB/Simulink is currently being developed to further extend this pedagogical approach.

Table 3. Integrated mapping of hands-on project stages, laboratory tasks, ABET learning outcomes, assessment criteria, and Kolb's experiential learning cycle [11] in a project-based Unit Operations Laboratory framework.

Hand-on project stage	Major tasks and activities	ABET LO	Assessment (Weight)
a) Pre-experimental phase	A1) Training in Aspen BatchSep under rating mode (rigorous design) by configuring key process parameters based on the conditions specified in **Table 1** A2) Formulate and define the project problem through literature survey and conceptual analysis A3) Identify experimental objectives and design an appropriate laboratory methodology	1-4, 6	15% - Conceptual understanding - Written proposal quality - Experimental design rigor
b) Experimental Phase	B1) Revise and finalize the experimental plan based on feedback B2) Execute the experiment following established procedures and safety and HAZOP protocols B3) A detail of three experiments that will be carried out in the lab B4) Perform preliminary data collection and on-site analysis B5) Demonstrate professionalism and adherence to laboratory practices	4, 5, 6	5% - Laboratory performance - Professional conduct
c) Post-experimental phase	C1) Conduct detailed data analysis and interpret experimental results C2) Compare experimental findings with theoretical predictions or simulation results C3) Evaluate system performance and identify sources of deviation or uncertainty C4) Prepare a comprehensive technical report with critical discussion	1, 3-6	65% - Analytical depth and technical accuracy - Quality of written report
d) Presentation and reflection analysis	D1) Deliver an oral presentation of the project findings D2) Reflect on learning outcomes, experimental challenges, and process improvements	6, 7	15% - Oral communication - Reflective writing

Table 4. Results of the survey on the digital twin-oriented laboratory using Aspen BatchSep and MATLAB/Simulink across the 2020-2023 academic years (n = 65). Scale is ranged from 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree and 5 = strongly agree.

No.	Questions	1	2	3	4	5	Mean	Variance	SD
1	The use of software enhances my ability to recognize errors compared to traditional instruction without software.	0	0	3	32	30	4.42	0.3352	0.58
2	The use of software tools contributes to a deeper understanding of my mistakes.	0	0	1	14	50	4.75	0.2163	0.47
3	Software applications (e.g., MATLAB/Simulink) improve my comprehension of fundamental concepts.	0	0	3	25	37	4.52	0.3418	0.58
4	I demonstrate greater engagement in class activities when software tools (e.g., Aspen BatchSep) are incorporated.	0	0	0	12	53	4.82	0.1505	0.39
5	I apply more critical thinking in class sessions that integrate software compared to those that do not.	0	0	0	6	59	4.91	0.0838	0.29
6	Integrating software tools into other courses would further improve my conceptual understanding in those subjects.	0	0	0	10	55	4.85	0.1302	0.36

5. CONCLUSIONS

This work demonstrates how a conventional batch distillation laboratory can be transformed into a digital-twin-oriented learning environment that connects thermodynamics, dynamic simulation, instrumentation, control, and experimental validation. By integrating a pilot-scale MeOH/H_2O batch distillation column with Aspen BatchSep, MATLAB/Simulink, LabVIEW HMI/SCADA, Node-RED, Modbus RTU RS485, MQTT, and OPC-UA communication, the proposed framework moves beyond the use of simulation as a passive pre-laboratory tool. Instead, students engage with a bidirectional learning workflow in which physical data, model predictions, control actions, and reflective analysis are continuously linked.

The educational value of this approach lies in its ability to make abstract separation and control concepts observable. Students do not only calculate vapor–liquid equilibrium behavior, reflux policies, and composition profiles; they also compare those predictions with real process trends, including tray temperatures, distillate composition drift, and deviations caused by heat loss, tray efficiency, and measurement uncertainty. The agreement between Aspen BatchSep predictions and experimental data for MeOH/H_2O separation provided a practical basis for teaching model validation, while the observed deviations became productive opportunities for engineering diagnosis rather than simple experimental error.

The framework also supports modern engineering-tool competency aligned with outcomes expected in contemporary chemical engineering education. Students practiced P&ID interpretation, signal tracing, data acquisition, control strategy formulation, simulation-based prediction, experimental execution, and technical reporting within a single project-based structure. Survey results across three academic years further suggest that students perceived the digital-twin-enhanced laboratory as improving conceptual understanding, error recognition, engagement, and critical thinking.

Although the present implementation focused on batch distillation of a binary MeOH/H_2O system, the framework is transferable to other unit operations and separation processes. Future extensions may include more rigorous online state estimation, advanced control, and additional case studies such as multicomponent or supercritical separation. Overall, this study provides a reproducible model for embedding digital-twin pedagogy into chemical engineering laboratories while preserving hands-on experimental authenticity.

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SUPPLEMENTARY DATA

https://drive.google.com/drive/folders/1rEavzxHH0wzwSaIr5_CUtjkCkCtDWbM5?usp=sharing

ABBREAVIATIONS

AAD%	Average Absolute Deviation percentage
ABET	Accreditation Board for Engineering and Technology
AI	Artificial intelligence; also used as Analog input in the WISCO I/O context
API	Application Programming Interface
BD	Batch distillation
ChE	Chemical Engineering
CSV	Comma-separated values
D	Distillate flow rate
DoE	Design of Experiments
DT	Digital twin
H_2O	Water
HAZOP	Hazard and Operability study
HETP	Height Equivalent to a Theoretical Plate
HMI	Human-Machine Interface
I/O	Input/output

ISA	International Society of Automation
JSON	JavaScript Object Notation
K-type TC	K-type thermocouple
Kp	Proportional gain of a PID controller
L	Liquid reflux flow rate
L/D	Liquid reflux-to-distillate ratio
LAH	Level Alarm High
LAL	Level Alarm Low
LO	Learning outcome
MATLAB	Matrix Laboratory software environment used with Simulink
MeOH	Methanol
Modbus RTU	Modbus Remote Terminal Unit serial communication protocol
MPC	Model Predictive Control
MQTT	Message Queuing Telemetry Transport
NRTL	Non-Random Two-Liquid activity-coefficient model
OPC-UA	Open Platform Communications Unified Architecture
P&ID	Piping and Instrumentation Diagram
PBL	Problem-Based Learning
PID	Proportional-Integral-Derivative control
PPE	Personal Protective Equipment
PSE	Process Systems Engineering
R	Reflux ratio
REST	Representational State Transfer
RS485	Recommended Standard 485 serial communication interface
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisition
SD	Standard deviation
SO	Student outcome
SS304L	Stainless steel grade 304L
TC	Thermocouple
TCP	Transmission Control Protocol
Td	Derivative time of a PID controller
Ti	Integral time of a PID controller
TIChE	Thai Institute of Chemical Engineering and Applied Chemistry Conference
Txy	Temperature-composition phase diagram
UNIQUAC	Universal Quasichemical activity-coefficient model
USB	Universal Serial Bus
VI	Virtual Instrument
VLE	Vapor-liquid equilibrium
WISCO I/O	WISCO input/output module and Analog (AI) Utilities configuration environment

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

In the preparation of this manuscript, the authors utilized ChatGPT (OpenAI) exclusively as a language support tool for grammar correction, stylistic refinement, and readability enhancement. All AI-generated suggestions were critically evaluated and edited by the authors, who retain full responsibility for the scientific validity and overall integrity of the manuscript.

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